Can you trust a KV cache you did not compute?

kv-notary · forging, and auditing, transformer KV caches column by column · github.com/PS12007/kv-notary · interactive demo

Abstract. Disaggregated serving, prefix caches and KV stores pass a transformer's KV cache between machines, and the reader trusts it. We show that this trust is easy to abuse and cheap to check. Forging a single column of the cache (one token's keys and values, computed from a nudged residual stream) makes Llama-3.2-1B give the attacker's answer in 86% of attempts, from any layer and almost any token position. Yet every cache is a deterministic function of its tokens, and each column can be recomputed against the claimed cache for the cost of one token. In 99.8% of successful forgeries this flags exactly the forged column, and no honest cache was flagged. Choosing columns by the question's value-weighted attention catches 91% of forgeries with 8 columns (Qwen2.5-0.5B: 77%), against 15% for 64 random columns, and 100% with 32 columns at 2044 tokens. Attacks that try to hide inside the bf16 tolerance usually fail or leave many columns failing, but one in 12 Qwen tasks fell to a forgery that stayed entirely inside it, so the tolerance itself is an attack surface that only deterministic kernels can close. A white-box attacker that knows the scoring rule can push its column down the ranking, Hidden forgeries often reach the question through later columns that read them, so we also trace the audit: follow what the most-read columns themselves read, using attention rows the recomputation yields for free. In a game of 8 attackers against 10 audit plans, tracing is the only plan that is best or tied-best on both models (the attacker's best reply wins 38% and 46% of the time, against 88% and 83% for the default score). No deterministic plan is a guarantee, and random columns remain the only attacker-independent one. Code, data and an interactive demo are public.

1. Introduction

A transformer's KV cache is a deterministic function of its tokens. Serving systems have started to treat it as data that moves. Disaggregated inference runs prefill on one pool of GPUs and ships the cache to another for decoding (DistServe, Splitwise, Mooncake). Prefix caches reuse one document's cache across users and requests (vLLM, SGLang). Cache stores such as LMCache persist caches to CPU memory, disk and remote storage, and resume sessions from them. In every one of these paths, the node that finally reads the cache trusts that it encodes the tokens it arrived with. Nothing checks.

That trust is worth testing, because the cache is far larger than the text and far less legible. A 450-token memo is 14 MiB of bf16 keys and values in Llama-3.2-1B. Any of it can be changed without changing a single token of the prompt, and no log, prompt filter or human reviewer would see the difference. This report asks three questions on two small open models:

1. **How little of a cache must change to change an answer?** One column, at any depth (§4).
2. **Can the reader check a cache without recomputing it?** Yes. Recomputing a column against the *claimed* cache costs one token through the model, and a cache is honest if and only if every

column passes (§3). Auditing the columns the question reads most catches non-adaptive forgeries with a few columns (§5).
3. **Where does the check break?** Against a white-box attacker who knows how audit columns are chosen (§6), and, rarely, inside the bf16 tolerance itself (§7).

2. Threat model

A **trusted node** holds the token ids of a document (a few bytes per token) and receives the document's KV cache from somewhere else. It then appends a question, decodes an answer, and returns it. The adversary controls the cache, and nothing else. We consider four adversaries:

Adversary	Realistic as	What it can do
Bit flip	silent data corruption in HBM, host RAM, NVMe, network	one random bit of one bf16 element
In-place edit	write access to a cache store or the wire	arbitrary changes to stored K/V; nothing downstream is recomputed
Consistent forger	a compromised or dishonest prefill worker	perturbs the residual stream of chosen columns *during* prefill and computes everything else honestly
Adaptive forger	the same, white-box, knows the audit rule	also optimises to keep its columns out of the audit

The attacker's goal is targeted: make the trusted node answer a specific wrong value (another city, project, colour or 4-digit code) for a question it anticipates. Cryptographic MACs on caches stop the first two adversaries and do nothing about the last two, because a dishonest prefill worker would sign whatever it computed.

3. Method

3.1 Column recomputation

Write the claimed cache as columns $c_0 \dots c_{T-1}$, where column p holds the post-RoPE keys and the values of token p at every layer. In an honest cache, $c_p = f(x_p, c_{<p})$: the model's computation for token x_p at position p, attending to the earlier columns. The verifier evaluates exactly this f, using the claimed $c_{<p}$ as context. At layer l, the recomputed query of token p attends to the claimed keys and values of positions $< p$ plus its own recomputed entry, so the cost is one token through the model plus one attention read of the cache. Any set of columns batches into one pass.

Soundness by induction. If every column passes, then c_0 is honest (it depends on x_0 alone). If $c_{<p}$ is honest and $c_p = f(x_p, c_{<p})$, so is c_p. So every tampered cache contains at least one failing column. The audit's job is to find one.

Two kinds of tampering look different. An in-place edit to column j at layer l also makes every later column that reads j above layer l disagree with its own recomputation, so it fails in many places. A consistent forger recomputes everything downstream of its change, so *only the forged columns fail*. That makes it the hard case.

3.2 Tolerance

In fp32, recomputed columns match the prefill to ~10^{-6} relative error. In bf16 they do not: the prefill's batched kernels and the verifier's single-column computation round differently. The verifier therefore compares the per-layer relative error $\|\hat{c} - c\|/\|c\|$ (K and V, all heads) to a threshold τ_l, set to 1.5× the largest honest error seen on 20 calibration documents. A column's score is $\max_l \text{error}_l/\tau_l$, and it fails above 1.

3.3 Choosing columns

Uniform sampling of k columns finds a single forged column with probability k/T. The verifier can do better because a forged column only matters if the question *reads* it, and the trusted node computes those attention weights anyway while it processes the question. Pre-norm transformers produce value vectors of bounded norm, so $a_{l,h,r,j} \cdot \|v_{l,j}\|$ bounds what column j can add to reader r through head h at layer l. We score each column by this value-weighted attention, take the max over heads and sum over the reader tokens (question plus the decoded answer), and then either average over layers (*mean*) or normalise per layer and take the max (*layermax*). The audit always checks the first four columns (attention sinks), then the top-ranked columns, then optionally some uniformly random ones. We also test a gradient score, $|\partial\text{NLL}/\partial K \cdot K| + |\partial\text{NLL}/\partial V \cdot V|$, and an ensemble that takes the union of the top columns under all three.

3.4 Attacks

All attacks optimise the teacher-forced likelihood of the target answer with Adam, through the trusted node's question pass. The **forger** adds a vector δ to the residual stream entering layer l at the chosen positions, with $\|\delta\| \le \beta\|h\|$, and backpropagates through the entire prefill, so the forged cache is consistent by construction. The **editor** optimises additive changes to stored K/V directly. The **adaptive forger** also pays a hinge penalty whenever a forged column's score is above 0.7× the score of the k-th ranked other column, computed with a differentiable copy of the audit score on the forged cache.

3.5 Setup

Models: Llama-3.2-1B-Instruct (16 layers, 8 KV heads) and Qwen2.5-0.5B-Instruct (24 layers, 2 KV heads), bf16, run by our own ~280-line implementation whose logits match `transformers` to 10^{-4} in fp32 (tested in CI on random tiny models). Documents: generated memos of about 450 tokens with ten facts in filler prose. We also use long versions with the facts scattered through Moby-Dick (about 2K tokens). We keep only tasks the model answers correctly from an honest cache, and set the target to the honest output with the value swapped. One RTX 5060 Laptop GPU (8 GB).

4. One column is enough

We forge one column, at one of five depths and three kinds of position: the last token of the fact's value, a uniformly random non-sink token, and the last document token. The forger may change the residual stream there by up to β times its norm. It then runs the rest of the prefill honestly. Over 24 memos per model (mean length 446 and 429 tokens) and 576 attacks per model, 446 of the Llama forgeries and 434 of the Qwen forgeries made the trusted node answer exactly the attacker's value. At β = 1, the success rate across all layers and positions is 86% for Llama and 80% for Qwen. Position barely matters: a forged column on an unrelated token, far from the fact, works about as well as one on the fact itself. The attack does not need to edit the fact. It needs one place the question reads.

The full column check confirms the structure from §3.1. In 99.8% (Llama) and 98.6% (Qwen) of successful forgeries, the forged column is the *only* column that fails. The few exceptions are forgeries that also change later columns enough to show. No honest cache failed a single column: the largest honest score over all 48 evaluation memos was 0.79 for Llama and 0.98 for Qwen, against a threshold of 1. These memos are disjoint from the 20 calibration memos.

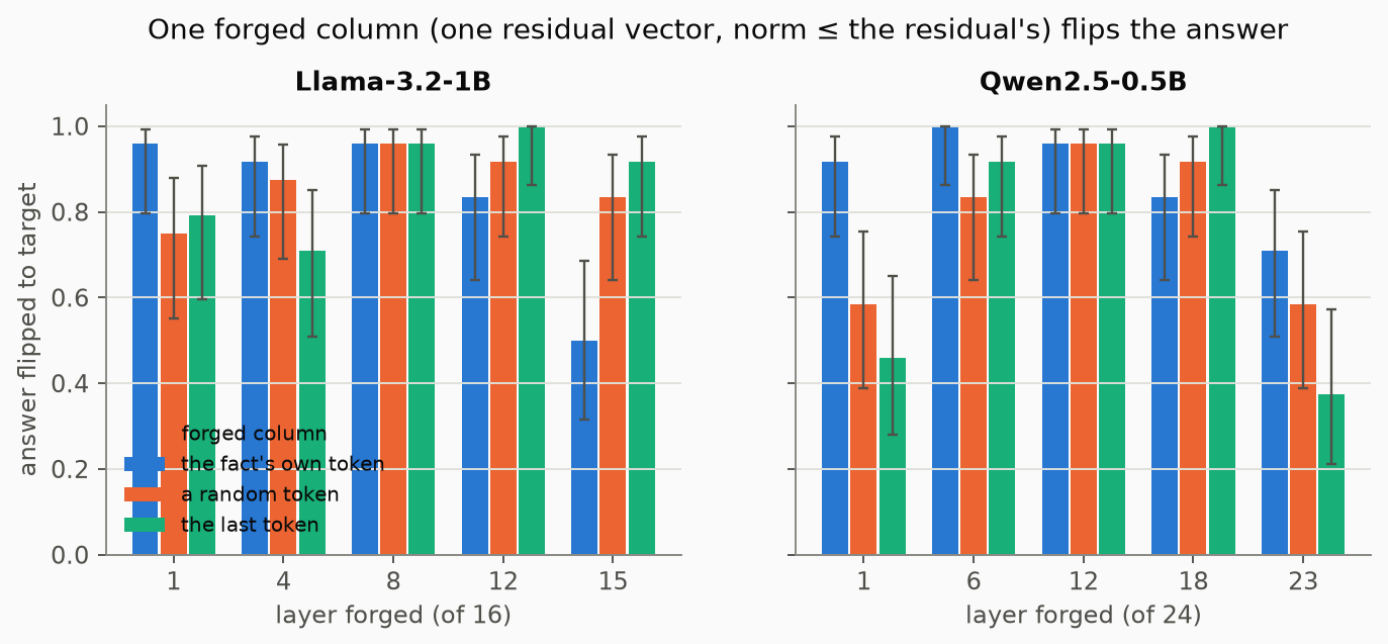


Figure 1. Success rate of single-column forgeries (β = 1) by layer and position kind, with 95% Wilson intervals.

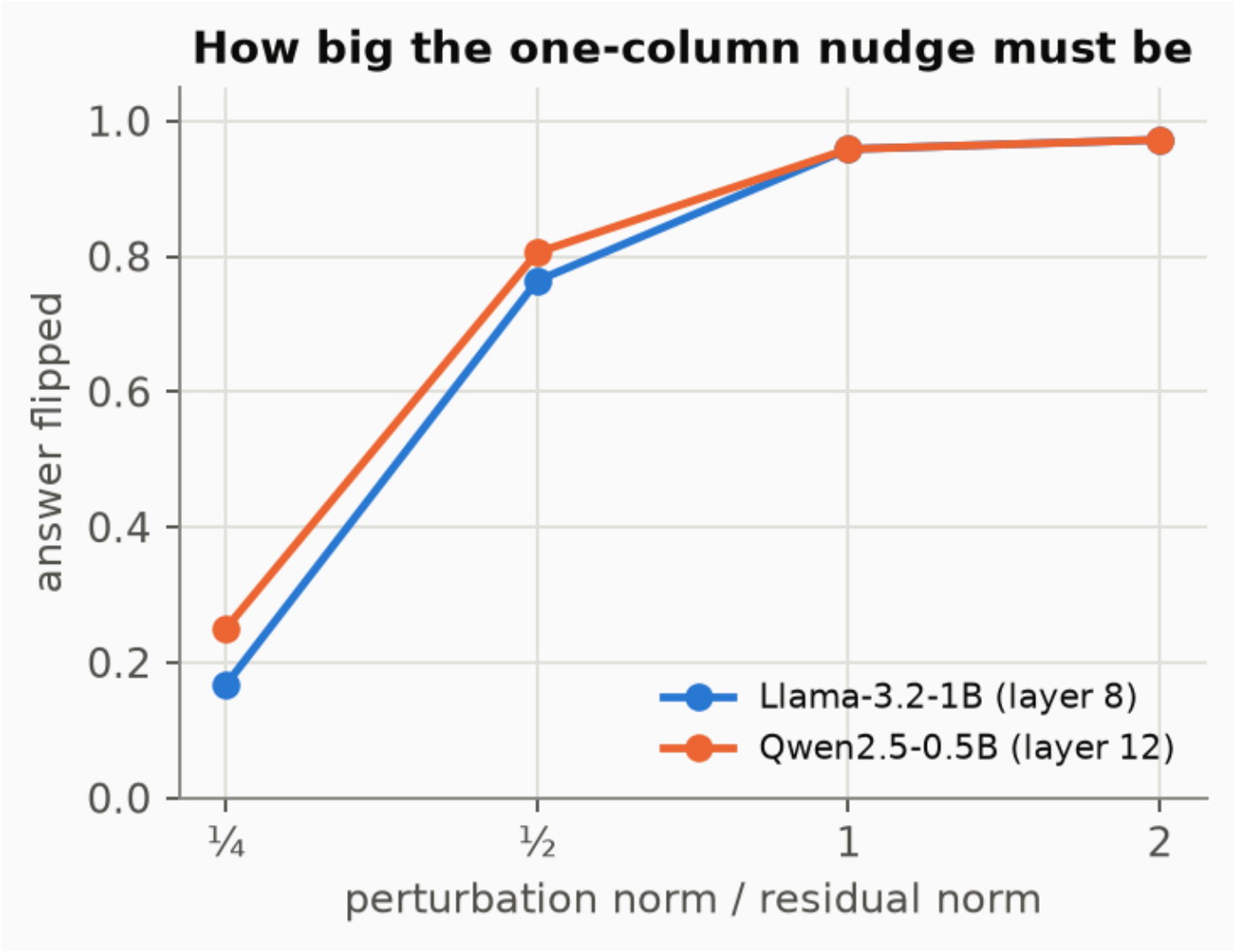


Figure 2. Success against the perturbation budget β at the middle layer.

model	layer	β	success	value pos.	random pos.	last pos.	only forged col. fails
Llama-3.2-1B	1	1	60/72	96%	75%	79%	98%
Llama-3.2-1B	4	1	60/72	92%	88%	71%	100%
Llama-3.2-1B	8	0.25	12/72	29%	8%	12%	100%
Llama-3.2-1B	8	0.5	55/72	71%	71%	88%	100%
Llama-3.2-1B	8	1	69/72	96%	96%	96%	100%
Llama-3.2-1B	8	2	70/72	92%	100%	100%	100%
Llama-3.2-1B	12	1	66/72	83%	92%	100%	100%
Llama-3.2-1B	15	1	54/72	50%	83%	92%	100%
Qwen2.5-0.5B	1	1	47/72	92%	58%	46%	96%
Qwen2.5-0.5B	6	1	66/72	100%	83%	92%	95%
Qwen2.5-0.5B	12	0.25	18/72	25%	21%	29%	100%
Qwen2.5-0.5B	12	0.5	58/72	83%	67%	92%	100%
Qwen2.5-0.5B	12	1	69/72	96%	96%	96%	100%
Qwen2.5-0.5B	12	2	70/72	100%	96%	96%	99%
Qwen2.5-0.5B	18	1	66/72	83%	92%	100%	100%
Qwen2.5-0.5B	23	1	40/72	71%	58%	38%	100%

Table 1. Single-column forgeries. 24 tasks × 3 positions per row.

5. Auditing where the question reads

A full check is a second prefill, so the practical question is how few columns an audit needs. Uniform sampling of k columns catches one bad column with probability k/T. The data match: 64 random columns catch 15% of Llama forgeries and 15% of Qwen forgeries. Checking the 4 sinks plus the 4 columns with the highest value-weighted attention (k = 8 in total) catches 91% and 77%. With k = 32 it catches 98% and 95%. The median forged column is the 5th most-read column, counting the four sinks: to change the answer, the forged column has to be read, and reading is visible. In 79% of Llama forgeries and 66% of Qwen forgeries, the single most-read non-sink column *is* the forged one.

Forgeries on the last document token are the easiest to find (top-8 catches 97% on Llama). The last token is read heavily by the question in any case. Forgeries on the fact's own value token are the hardest (84%), because the honest value token is already among the most-read, so a forgery there competes with the real signal. Qwen is harder throughout. With only 2 KV heads, its attention spreads over more columns, so the forged one is less of an outlier.

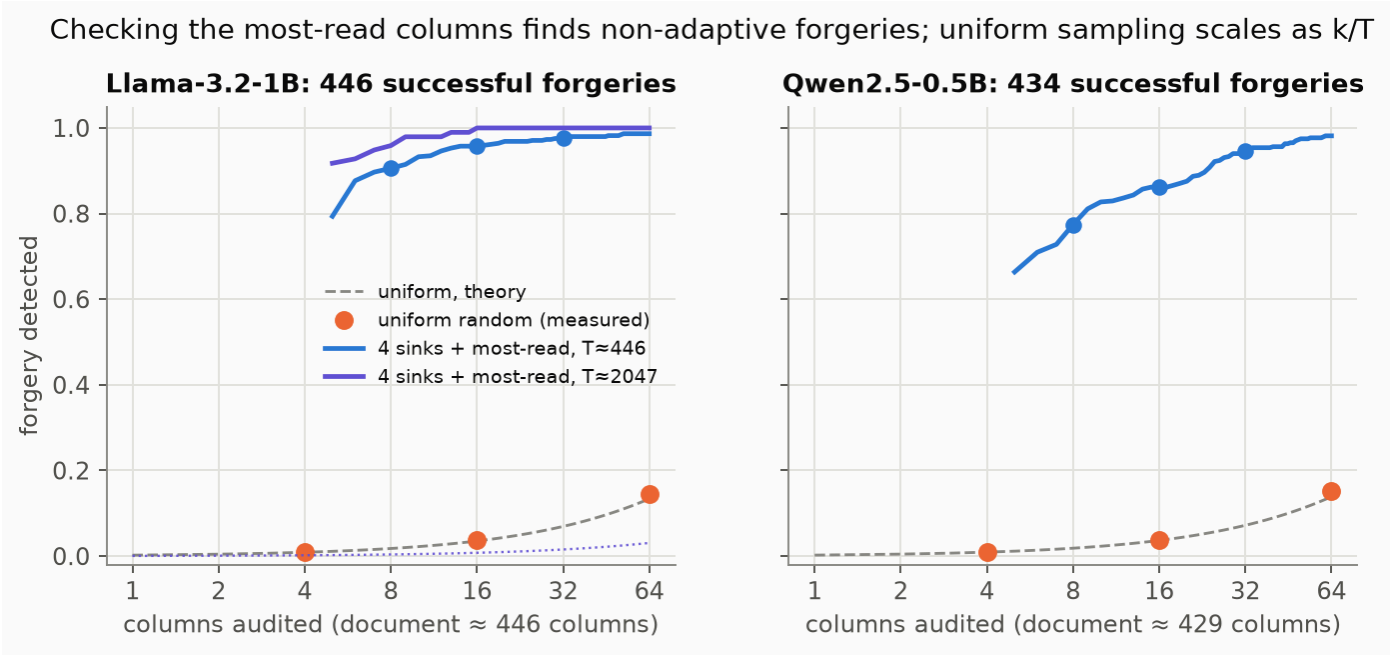


Figure 3. Probability that an audit of k columns contains the forged column. Exact hypergeometric probabilities for random plans, averaged over successful forgeries.

model	forgeries	top-8	top-16	top-32	top-8 + 8 random	random 4	random 16	random 64
Llama-3.2-1B	446	91%	96%	98%	91%	0.9%	3.7%	14.5%
Qwen2.5-0.5B	434	77%	86%	95%	78%	0.9%	3.8%	15.1%

Table 2. Detection of successful single-column forgeries. Top-k includes the 4 sink columns.

5.1 Long documents

The ratio matters more as documents grow. With the facts scattered through Moby-Dick (Llama, layers L/4, L/2, 3L/4), a 2044-token cache leaves uniform 64-column audits at 3.2%, while top-32 still catches 100%. The forged column's median rank stays small as T grows, so the audit cost in columns stays flat while the uniform cost grows linearly.

T (tokens)	success	only forged fails	top-8	top-16	top-32	random 64	median rank
2044	97/108	99.0%	96%	100%	100%	3.2%	5

Table 3. Llama-3.2-1B on long documents.

6. Adaptive attackers

The audit's strength in §5 rests on a forged column having to be read. An attacker who knows the scoring rule can try to be read *enough* without being read *most*. Our adaptive forger adds a hinge penalty that pushes each forged column's score below 0.7× the score of the k-th ranked other column. The score is computed on its own forged cache with the trusted node's question, which we let the attacker know in advance. It attacks one of two scores: *mean* (the default in §5) or *layermax* (per-layer normalised, max over layers). It can also spread the edit over m random columns.

Llama-3.2-1B

attacker	success	rank (mean)	rank (layermax)	rank (grad)	top-32 mean	top-32 layermax	ensemble-16	top-32 + 32 random	random 64	direct
no hiding (λ = 0)	16/16	0	0	0	100%	100%	100%	100%	14%	100%
m = 1, hide below top-4, vs. mean	14/16	10	0	22	100%	100%	100%	100%	15%	71%
m = 1, hide below top-28, vs. mean	10/16	50	11	49	0%	100%	70%	8%	15%	60%
m = 1, hide below top-4, vs. layermax	12/16	38	10	54	42%	100%	92%	47%	15%	83%
m = 1, hide below top-12, vs. layermax	9/16	20	23	38	67%	78%	56%	69%	16%	78%
m = 1, hide below top-28, vs. layermax	8/16	50	35	47	25%	38%	38%	32%	16%	75%
m = 1, hide below top-60, vs. layermax	6/16	2	8	26	83%	83%	83%	85%	14%	100%
m = 4, hide below top-12, vs. layermax	13/16	4	7	14	77%	100%	85%	84%	47%	92%
m = 16, hide below top-12, vs. layermax	13/16	0	0	1	100%	100%	100%	100%	92%	92%
m = 64, hide below top-12, vs. layermax	15/16	0	0	0	100%	100%	100%	100%	100%	100%

Qwen2.5-0.5B

attacker	success	rank (mean)	rank (layermax)	rank (grad)	top-32 mean	top-32 layermax	ensemble-16	top-32 + 32 random	random 64	direct
no hiding (λ = 0)	16/16	0	0	7	100%	100%	100%	100%	18%	100%
m = 1, hide below top-4, vs. mean	15/16	12	0	23	87%	87%	87%	88%	16%	73%
m = 1, hide below top-28, vs. mean	12/16	49	1	100	8%	92%	75%	17%	17%	50%
m = 1, hide below top-4, vs. layermax	15/16	24	11	44	53%	67%	67%	58%	20%	40%
m = 1, hide below top-12, vs. layermax	11/16	24	21	56	55%	82%	45%	60%	17%	73%
m = 1, hide below top-28, vs. layermax	9/16	87	51	110	33%	0%	0%	39%	15%	56%
m = 1, hide below top-60, vs. layermax	6/16	46	90	72	33%	17%	33%	40%	17%	33%
m = 4, hide below top-12, vs. layermax	13/16	10	14	28	77%	92%	77%	84%	51%	92%
m = 16, hide below top-12, vs. layermax	14/16	6	12	11	100%	100%	100%	100%	93%	100%
m = 64, hide below top-12, vs. layermax	14/16	0	0	1	100%	100%	100%	100%	100%	93%

Table 4. Adaptive forgeries at the middle layer, β = 1. Ranks are the median over successful attacks of the best-hidden forged column's rank among non-sink columns (0 = most read). Detection columns are the probability of catching it. Ensemble-16 is the union of the top 16 under all three scores, plus the sinks (≈ 30–45 columns). *Direct* is the fraction where putting the honest forged columns back restores the honest answer, so the effect does not run through other columns.

A single score can be gamed. Without the penalty, the forged column is ranked 0 (median, among non-sink columns). An attacker hiding below the top 28 under the mean score still succeeds in 10/16 tasks and lands at rank 50, so top-32 under that score catches it 0% of the time. The same forgery is ranked 11 by layermax: the mean score is low because the column is read hard at a few layers, not lightly everywhere.

Gaming the stronger score works too. Attacking layermax directly with hide-28, the forger succeeds in 8/16 tasks. Its column then ranks 35 under layermax, 50 under mean and 47 under the gradient score, and top-32 layermax catches 38%. The union of the top 16 under all three scores catches 38%, with about 32 columns. This is the honest result: an attacker that knows the score can lower it. A deterministic top-k audit is an arms race, not a guarantee.

Qwen2.5-0.5B is easier to game. The same attacker succeeds in 9/16 tasks, and top-32 layermax catches 0% of them, the ensemble 0%. Only the random columns still find it, at the k/T rate (15% for 64).

Spreading out costs the attacker detectability elsewhere. With m = 64 forged columns hidden from layermax top-12, it succeeds in 15/16 tasks, but each of those caches has 64 failing columns, so 64 random columns catch it 100% of the time. That is the case random draws are for. Mixing the two (top-32 + 32 random) covers both ends: 100% here.

So the deterministic part of an audit should be treated as a cheap filter for non-adaptive faults and attacks. Guarantees against an adaptive forger come only from the random part, at k/T per forged column, or from a full recompute of whatever the audit's budget cannot cover.

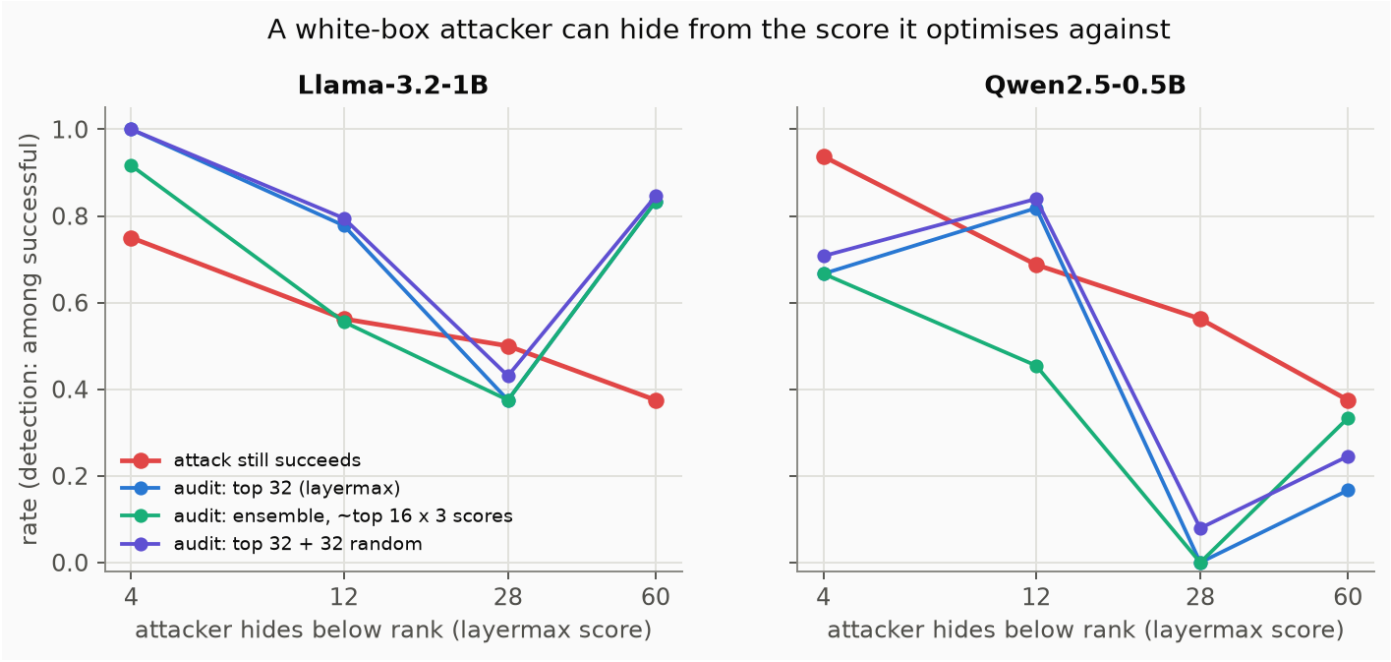


Figure 4. Adaptive attackers. Left: where the forged column lands in each ranking. Right: detection by audit plan.

6.1 Round two: audits the attacker cannot see

Two things stand out in Table 4. Different scores disagree about where a hidden column lands, and in 24% (Llama-3.2-1B) and 44% (Qwen2.5-0.5B) of successful hidden single-column forgeries, putting the honest column back does *not* restore the answer. The forgery has been *laundered*: later columns read the forged one during prefill, carry its effect forward, and the question reads them instead. Those later columns are consistent with their inputs, so they pass the check. This suggests two defenses. The first is scores the attacker cannot optimise against: finer ones (per head, per reader token) and a **secret** one, a random quarter of the heads with random weights, drawn by the verifier for each audit. The second is **tracing**. The audit recomputes its columns anyway, and the recomputation yields each column's own attention to earlier columns. So after auditing the most-read columns, the verifier follows what *they* read most (value-weighted, per head, max over layers), audits those, and repeats until the budget is spent (Algorithm: half the budget on merged public scores, the rest in rounds of B/8 traced columns).

We test this against seven attackers that hide one forged column below the 28 most-read columns of one public score, of all four at once, or of all four *and* the tracing rule itself, using a differentiable copy of the trace rows on the forged prefill. 24 tasks per model, 150 Adam steps, middle layer, β = 1. Every plan audits exactly 32 columns, including the 4 sinks.

Llama-3.2-1B

attacker	success	mean	layermax	headmax	tokmax	gradient	all public merged	secret heads	trace	trace + random	random
no hiding	24/24	100%	100%	100%	100%	100%	100%	99%	100%	100%	7%
hides from mean	21/24	0%	95%	95%	100%	14%	100%	47%	95%	50%	7%
hides from layermax	17/24	35%	12%	59%	88%	12%	76%	48%	53%	38%	7%
hides from headmax	17/24	71%	82%	12%	82%	41%	76%	49%	47%	27%	7%
hides from tokmax	13/24	92%	92%	92%	69%	92%	92%	86%	85%	85%	7%
hides from all public	9/24	0%	0%	0%	0%	0%	0%	12%	44%	36%	8%
hides from all public + trace	7/24	0%	0%	0%	14%	14%	14%	11%	29%	31%	8%
hides from all public + exact trace	5/24	0%	0%	0%	0%	0%	0%	15%	20%	4%	8%

Qwen2.5-0.5B

attacker	success	mean	layermax	headmax	tokmax	gradient	all public merged	secret heads	trace	trace + random	random
no hiding	24/24	100%	100%	100%	100%	100%	100%	89%	100%	100%	9%
hides from mean	21/24	5%	95%	81%	62%	14%	67%	37%	90%	46%	13%
hides from layermax	18/24	56%	6%	17%	33%	11%	6%	30%	39%	24%	13%
hides from headmax	18/24	56%	50%	0%	28%	6%	17%	28%	50%	20%	10%
hides from tokmax	14/24	71%	86%	71%	0%	50%	71%	61%	86%	39%	9%
hides from all public	11/24	0%	0%	0%	0%	9%	0%	9%	55%	15%	11%
hides from all public + trace	9/24	0%	0%	0%	0%	0%	0%	10%	44%	16%	11%
hides from all public + exact trace	6/24	17%	0%	0%	0%	0%	0%	16%	0%	4%	8%

Table 5. Probability that a 32-column audit catches a successful forgery, by attacker (rows) and audit plan (columns). Secret: mean over 32 secret draws. Random parts are exact hypergeometric probabilities.

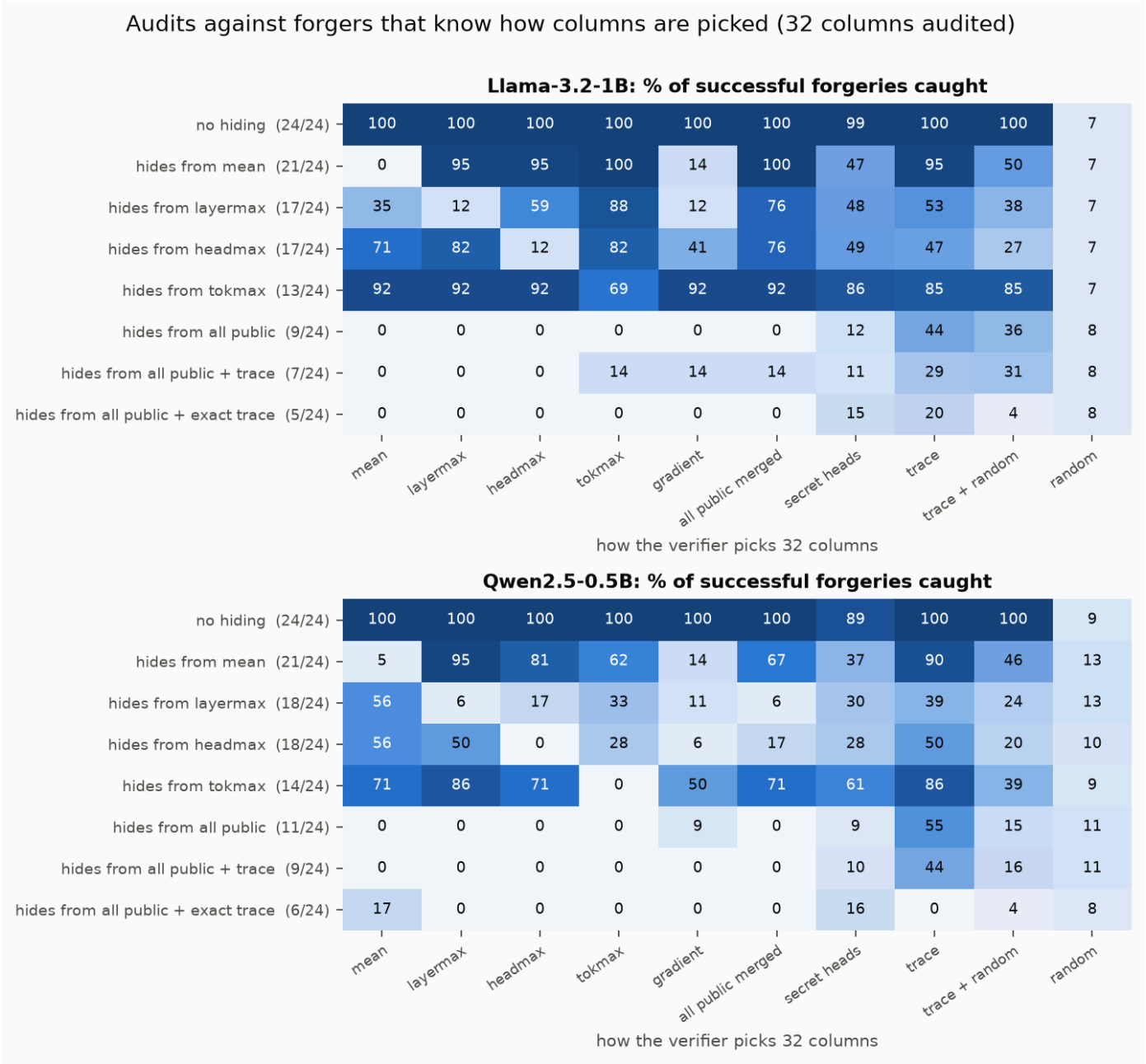


Figure 5. Rounds two and three of the arms race. Each cell: share of successful forgeries a 32-column audit catches.

The game. An audit plan is only as good as the attacker's best reply to it. For each plan, Table 6 gives the share of tasks in which the *best* of our attackers both flips the answer and escapes the audit (success rate × miss rate, maximised over attackers). On Llama-3.2-1B, against a 32-column audit the attacker's best reply wins 88% of the time if the verifier uses the default mean score, 93% against random columns, and 38% against the best plan, *tokmax* (its best reply there: *hides from all public*). On Qwen2.5-0.5B, against a 32-column audit the attacker's best reply wins 83% of the time if the verifier uses the default mean score, 91% against random columns, and 46% against the best plan, *trace* (its best reply there: *hides from layermax*).

audit plan	Llama B=16	Llama B=32	Llama B=64	Qwen2.5 B=16	Qwen2.5 B=32	Qwen2.5 B=64
mean	88%	88%	25%	88%	83%	29%
layermax	67%	62%	21%	75%	71%	25%
headmax	62%	62%	29%	75%	75%	46%
tokmax	38%	38%	21%	75%	58%	33%
gradient	88%	75%	42%	83%	75%	62%
all public merged	58%	38%	33%	75%	71%	38%
secret heads	60%	47%	29%	68%	55%	41%
trace	54%	38%	17%	62%	46%	25%
trace + random	97%	52%	34%	97%	60%	38%
random	96%	93%	85%	95%	91%	82%

Table 6. Worst-case attacker win rate per audit plan (lower is better for the verifier), at audit budgets of 16, 32 and 64 columns.

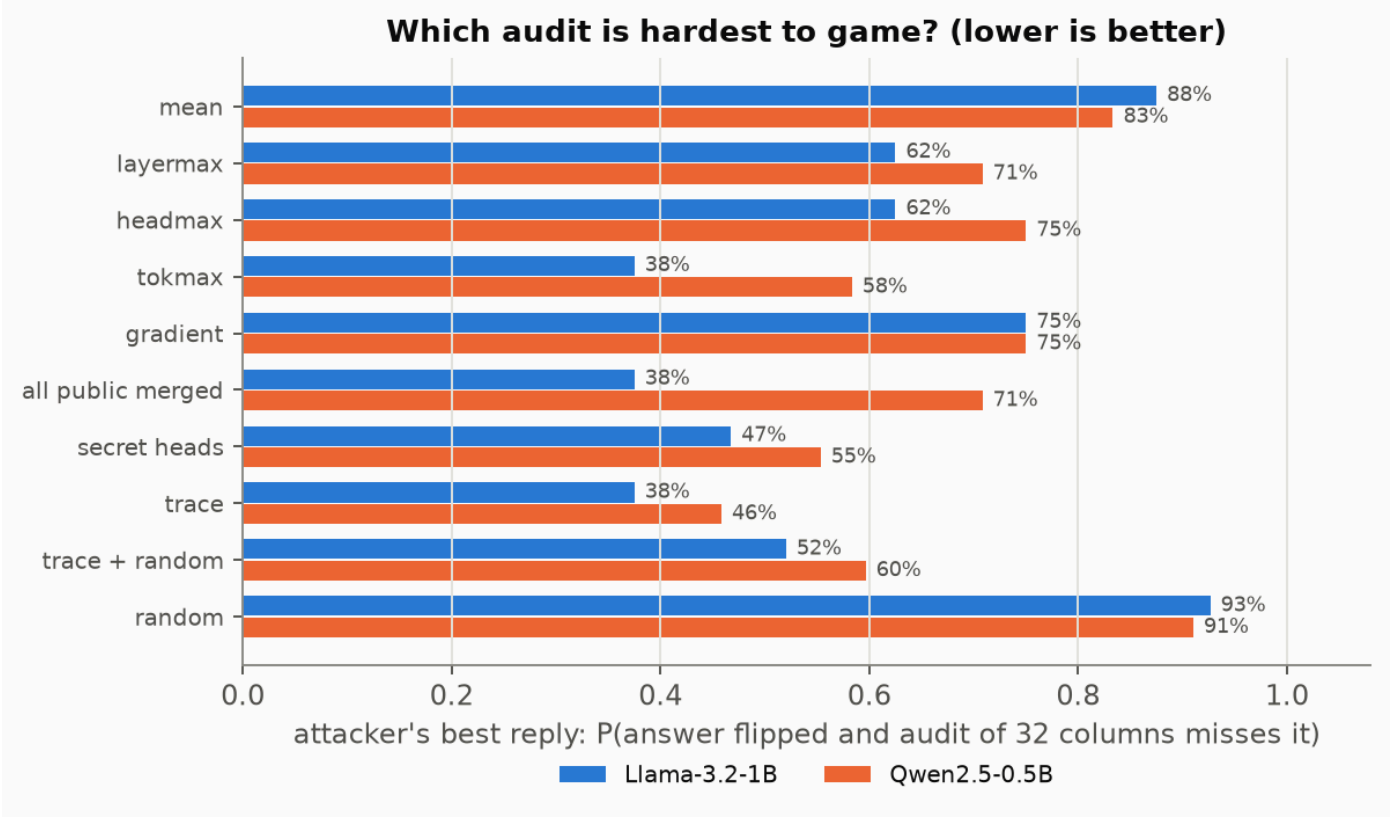


Figure 6. The attacker's best reply to each 32-column audit plan: how often it both flips the answer and escapes the audit.

What holds on both models. (1) The default mean score is the easiest to game: the best reply beats a 32-column mean audit 88% of the time on Llama-3.2-1B and 83% of the time on Qwen2.5-0.5B, no better than 32 random columns (93%, 91%). (2) No single public score is robust across models. The finest one, tokmax (per head, layer and reader token), holds the attacker to 38% on Llama-3.2-1B but 58% on Qwen2.5-0.5B. (3) Tracing is the most consistent plan: 38% on Llama-3.2-1B and 46% on Qwen2.5-0.5B. It does not rely on any one score, but follows the path the forged information has to take to reach the question. The attacker that also optimises against the trace rule did exactly as well as the one that ignores it, in success and in detection. Our differentiable copy of the trace covers one hop from a fixed set of read columns, while the verifier keeps following hops until its budget runs out; a stronger trace-aware attacker is

the obvious next step. (4) Secret head subsets help only modestly (47%, 55%): an attacker that is quiet in every head is quiet in any subset of them.

Two caveats. The attackers hide below rank 28, which matches the 32-column budget. At 64 columns every attention-based plan looks strong (tracing 17% on Llama-3.2-1B, tracing 25% on Qwen2.5-0.5B) only because the attackers were not told about the larger budget. And all of these are games against our optimiser (150 Adam steps, one hinge loss per score), so the rates describe a modest white-box attacker, not a bound. Random columns give the only attacker-independent guarantee, and at k/T it is a weak one.

Round three: an attacker that knows the trace exactly. We gave the forger a replay of the verifier's own trace plan (same budget, same rounds, same weights, on its own forged prefill), and a penalty whenever its column would be picked, on top of hiding from all four public scores. On Llama-3.2-1B, it succeeds in only 5/24 tasks; tracing catches 20% of those, so it beats the traced audit in 17% of tasks, below the best reply to tracing among the other attackers (38%). On Qwen2.5-0.5B, it succeeds in only 6/24 tasks; tracing catches 0% of those, so it beats the traced audit in 25% of tasks, below the best reply to tracing among the other attackers (46%). Knowing the rule lets the attacker evade it, but satisfying it costs most of its success. A plausible reading, which we have not tested directly: the forged information has to reach the question through some column the question reads, and keeping every hop of that path quiet leaves little room to change the answer. The worst case for tracing is still set by simpler attackers.

7. Hiding inside the tolerance

bf16 forces a tolerance, and a tolerance is room. We test two attackers that try to stay inside it. The *edit* attacker changes every stored entry at every layer in place, by at most $\rho \times$ the largest honest error. The *distributed forger* nudges the residual stream of every document column at the middle layer by a relative ε, then computes everything else honestly, so each column carries only its own small inconsistency.

model	dtype	attack	ρ or ε	success	failing columns (mean)	fewest failing when it worked	median max score
Llama-3.2-1B	bf16	edit	0.1	0%	0 / 446	–	0.9
Llama-3.2-1B	bf16	edit	0.3	58%	278 / 446	131	2.9
Llama-3.2-1B	bf16	edit	1	100%	444 / 446	427	8.1
Llama-3.2-1B	bf16	forge	0.001	0%	0 / 446	–	0.7
Llama-3.2-1B	bf16	forge	0.003	0%	0 / 446	–	0.8
Llama-3.2-1B	bf16	forge	0.01	8%	9 / 446	8	1.7
Llama-3.2-1B	bf16	forge	0.03	8%	298 / 446	290	4.9
Llama-3.2-1B	bf16	forge	0.1	100%	442 / 446	425	10.3
Llama-3.2-1B	fp32	edit	0.1	0%	0 / 444	–	0.9
Llama-3.2-1B	fp32	edit	0.3	0%	73 / 444	–	2.1
Llama-3.2-1B	fp32	edit	1	0%	436 / 444	–	6.9
Llama-3.2-1B	fp32	forge	1e-05	0%	437 / 444	–	7.3
Llama-3.2-1B	fp32	forge	0.0001	0%	440 / 444	–	72.4
Llama-3.2-1B	fp32	forge	0.001	0%	440 / 444	–	731.9
Llama-3.2-1B	fp32	forge	0.01	12%	440 / 444	446	6669.5
Llama-3.2-1B	fp32	forge	0.1	100%	440 / 444	425	43598.8
Qwen2.5-0.5B	bf16	edit	0.1	8%	127 / 430	104	2.4
Qwen2.5-0.5B	bf16	edit	0.3	83%	426 / 430	408	7.0
Qwen2.5-0.5B	bf16	edit	1	100%	429 / 430	410	20.1
Qwen2.5-0.5B	bf16	forge	0.001	0%	0 / 430	–	0.7
Qwen2.5-0.5B	bf16	forge	0.003	0%	0 / 430	–	0.7
Qwen2.5-0.5B	bf16	forge	0.01	17%	1 / 430	0	1.1
Qwen2.5-0.5B	bf16	forge	0.03	25%	75 / 430	44	2.7
Qwen2.5-0.5B	bf16	forge	0.1	100%	426 / 430	407	6.9

Table 7. Attacks that try to stay inside the bf16 (or fp32) tolerance.

Llama-3.2-1B bf16: the quietest successful attack (forge, $\varepsilon = 0.01$) left 8 of 446 columns failing, which 16 random columns would catch with probability 25% and 64 with 69%; Llama-3.2-1B fp32: the quietest successful attack (forge, $\varepsilon = 0.1$) left 425 of 444 columns failing, which 16 random columns would catch with probability 100% and 64 with 100%. Below the tolerance ($\rho = 0.1$, $\varepsilon \le 0.003$ in bf16) nothing succeeded on Llama. The band between the tolerance and an effective attack is narrow, because §4 needed perturbations on the order of the residual norm while the tolerance is a few percent of it, and spreading the change over 400 columns does not change that: attention averages the nudges, so effective spread attacks usually break many columns at once, which is exactly what random columns are good at. In fp32 the tolerance is $\sim 10^{-6}$ and every attack lit up almost every column.

But the tolerance is an attack surface. On Qwen2.5-0.5B (bf16), 1 of 12 tasks fell to a distributed forgery ($\varepsilon = 0.01$) that flipped the answer with *every* column inside the tolerance: its worst column scored just under 1. No column audit, however chosen, can catch that cache, because by the verifier's own standard it is honest. Tightening the threshold would catch it only at the price of false alarms on honest caches, whose worst score was 0.98 on this model. The real fix is a smaller tolerance: batch-invariant,

deterministic kernels in prefill and verification, or a verification precision the prefill also uses. In fp32 (tested on Llama), every forgery of this kind lit up hundreds of columns.

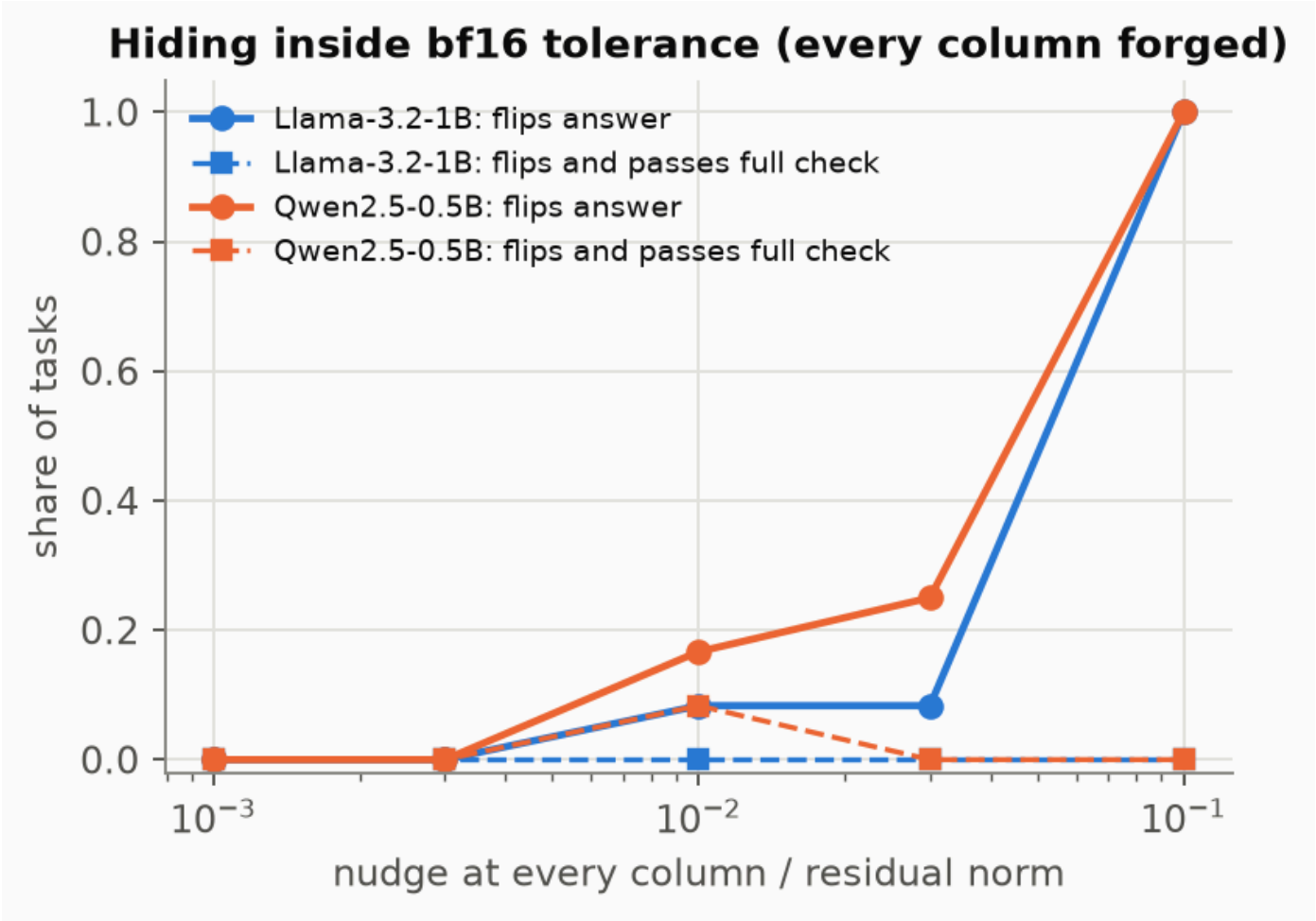


Figure 7. In-tolerance attacks: success rate and the number of failing columns as the per-column budget grows.

7.1 Across software stacks

A verifier will rarely run the same kernels as the prefill worker. We prefill with Hugging Face `transformers` (bf16, SDPA) and verify with kv-notary's own column recomputation (`kvnotary.hf` converts the cache). We score honest transformers caches with the thresholds calibrated on kv-notary's own caches, then with thresholds calibrated on transformers caches of the same 20 calibration documents. Then we forge one column with kv-notary's forward, as in §4, and decode it with transformers.

model	memos	false alarms, native τ	worst honest score	false alarms, τ from transformers	worst honest score	τ ratio, median (max)	forgeries that fool transformers	weakest forged score
Llama-3.2-1B	24	1/24	1.45	0/24	1.00	1.05 (1.45)	23/23	22.7
Qwen2.5-0.5B	24	24/24	4107.13	0/24	0.80	1.23 (6.16e+03)	23/23	14.2

Table 8. Prefill with transformers, verify with kv-notary. Scores are relative to the threshold (1 = at tolerance).

Three results. Forgeries transfer: every forged cache that fooled kv-notary's decoder also fooled transformers', so the attack does not depend on the implementation that consumes it. The tolerance must

be calibrated against the stack that produces the cache: kv-notary's first layer recomputes Qwen bit-exactly, so its native threshold there is the 10^{-6} floor, and every honest transformers cache fails it. Calibrated on transformers caches, the thresholds are mostly within a few tens of percent of the native ones, honest caches pass, and forged columns still score an order of magnitude above the threshold. Cross-stack verification works, but calibration is part of the deployment, not a constant.

7.2 One cache, many questions

The attacker targets one question, but a node that keeps a cache usually answers several. For every successful forgery we ask the memo's nine other questions and record whether their answers change (collateral damage), and whether a 32-column audit driven by each other question, or by all nine at once, would contain the forged column. All audits merge the four public scores.

model	attacker	success	other answers changed	forgeries with no collateral	caught: target question	caught: one other question	caught: all 9 other questions	caught: all 10 questions
Llama-3.2-1B	plain forger	24/24	63%	0%	100%	100%	100%	100%
Llama-3.2-1B	stealth forger (hides from target question)	9/24	17%	22%	22%	22%	11%	11%
Qwen2.5-0.5B	plain forger	23/24	44%	4%	96%	86%	100%	96%
Qwen2.5-0.5B	stealth forger (hides from target question)	17/24	27%	18%	0%	5%	6%	6%

Table 9. Collateral damage and audits driven by other questions (32 columns, public scores merged).

Plain forgeries are blunt. They change 63% and 44% of the other answers (Llama, Qwen), almost none leave every other answer intact (0% and 4%), and an audit driven by any single other question catches 100% and 86% of them. **Stealth forgeries are quieter but not clean.** They change 17% and 27% of the other answers, and only 22% and 18% leave every other answer intact. A user who asks several questions would often see something wrong, though the node itself cannot tell without knowing the honest answers. Yet audits driven by other questions catch only 22% and 5% of stealth forgeries (one question) and 11% and 6% (all nine at once, 32 columns). Hiding from the target question's attention hides from the other questions' attention too, even when their answers move, which fits the laundering seen in §6.1: the effect travels through honest-looking columns. An audit driven by other traffic is therefore no substitute for auditing the question at hand.

8. Naive tampering and corruption

In-place edits. An editor that optimises stored K/V directly at one column, without recomputing anything later, makes every later column that reads the edited one disagree with its own recomputation. That holds at every layer except the last, which no later layer reads. So these edits are loud, and even a single random column usually catches them.

model	edited layer	success	failing columns (median)	of later columns	random 1	random 4
Llama-3.2-1B	0	67%	216	100%	50%	84%
Llama-3.2-1B	4	100%	216	100%	50%	84%
Llama-3.2-1B	8	100%	216	100%	48%	84%
Llama-3.2-1B	12	96%	216	100%	49%	84%
Llama-3.2-1B	15	100%	1	0%	0%	1%
Qwen2.5-0.5B	0	0%	212	100%	51%	84%
Qwen2.5-0.5B	6	50%	212	100%	51%	84%
Qwen2.5-0.5B	12	83%	212	100%	51%	84%
Qwen2.5-0.5B	18	21%	186	91%	41%	77%
Qwen2.5-0.5B	23	8%	1	0%	0%	1%

Table 10. In-place edits of one column (24 per row: 12 tasks × 2 positions).

Bit flips. We flip 2304 random bits per model, uniformly over element, layer, head, position and bit. For Llama, 4.9% change the decoded tokens and 3.7% make the answer wrong. Nearly all the damage comes from bit 14, the top exponent bit, which turns a value of magnitude ~1 into one of ~10^{38}. A range check ($|x|$ above the largest honest magnitude at that layer) catches 100% of the harmful flips at zero cost. The full column check flags 28% of all flips, including harmless ones. Hardware faults are therefore the cheap case: a range check plus a small random audit covers them.

model	bit	flips	output changed	answer wrong	check flags it	range check fires
Llama-3.2-1B	15 (sign)	146	0%	0%	58%	0%
Llama-3.2-1B	14 (exp. MSB)	132	64%	61%	96%	79%
Llama-3.2-1B	13 (exp.)	147	2%	1%	36%	14%
Llama-3.2-1B	7 (exp. LSB)	158	0%	0%	27%	1%
Llama-3.2-1B	0 (mantissa LSB)	140	0%	0%	0%	0%
Qwen2.5-0.5B	15 (sign)	146	0%	0%	75%	0%
Qwen2.5-0.5B	14 (exp. MSB)	132	59%	56%	98%	75%
Qwen2.5-0.5B	13 (exp.)	147	1%	0%	42%	14%
Qwen2.5-0.5B	7 (exp. LSB)	158	0%	0%	37%	1%
Qwen2.5-0.5B	0 (mantissa LSB)	140	0%	0%	7%	0%

Table 11. Random single-bit flips by bit position (selected bits).

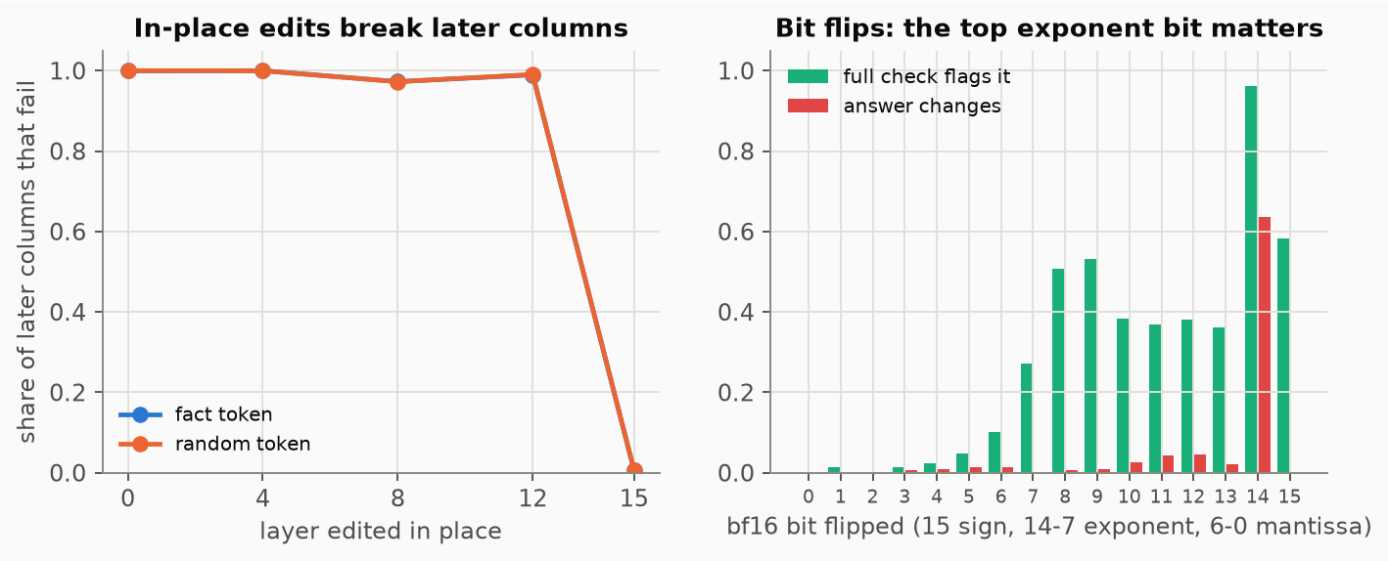


Figure 8. Left: columns that fail after an in-place edit, by edited layer. Right: effect of a single bit flip by bit position.

9. Cost

On NVIDIA GeForce RTX 5060 Laptop GPU, Llama-3.2-1B, bf16. The influence pass reads the cache once with the question tokens, which the trusted node does anyway to answer, so in a fused implementation it would be nearly free. Here we time it separately. An audit of 16 columns plus the influence pass costs 136.7% of a prefill at 512 tokens and 6.1% at 16384 tokens. The audit has a floor of about 30 ms: our unfused PyTorch code launches dozens of small kernels per layer, so on short documents it costs as much as a prefill. Above that floor, the prefill grows quadratically in T while the audit is a fixed number of tokens attending to T columns, so the gap widens with length. Our implementation is plain PyTorch with explicit attention for the audited rows. The prefill uses SDPA. Most of that floor is CPU time spent launching about 1,100 small kernels; the GPU work for 16 columns is about one read of the weights. Capturing the audit as a CUDA graph (`kvnotary.fast.GraphedAudit`, bit-identical output) cuts 16 columns at 512 tokens from 29 ms to 9 ms. A traced audit of 32 columns runs its recomputation in about five sequential rounds, so it pays the floor several times: 187 ms at 16384 tokens, 7.3% of the prefill. Fusing those rounds into one kernel launch each would bring it close to the plain audit.

T	KV (MiB)	prefill (ms)	influence (ms)	8 cols (ms)	16 cols (ms)	64 cols (ms)	16 cols, CUDA graph (ms)	trace 32 (ms)	full check (ms)	influence + 16 / prefill
512	16	43	30.1	30.7	28.9	28.8	8.7	133	66	136.7%
1024	32	81	30.1	28.1	28.2	29.0	11.9	161	197	71.6%
2048	64	181	44.3	27.9	28.1	27.4	12.6	131	667	40.1%
4096	128	429	36.4	31.5	28.5	42.1	14.3	153	2106	15.1%
8192	256	1003	61.0	33.6	29.0	71.6	–	131	–	9.0%
16384	512	2558	104.3	36.8	51.8	130.8	–	187	–	6.1%

Table 12. Median wall-clock time of 5 runs.

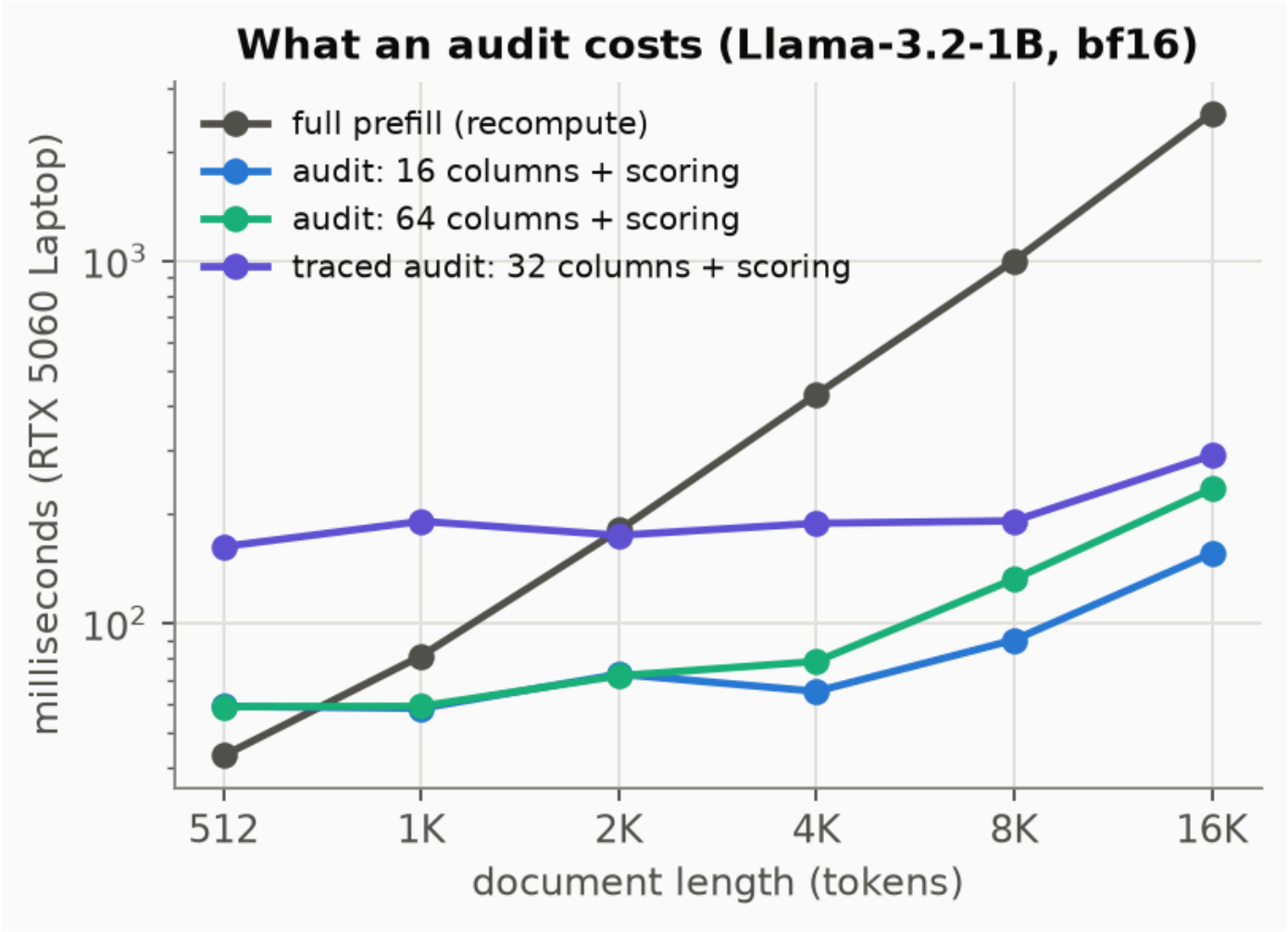


Figure 9. Time for prefill, a full check, and audits of 16 and 64 columns, by document length.

10. Related work

Moving KV caches. Disaggregated serving separates prefill from decode and transfers the cache between them (DistServe, Zhong et al. 2024; Splitwise, Patel et al. 2024; Mooncake, Qin et al. 2024). Prefix caching reuses caches across requests (vLLM's automatic prefix caching, Kwon et al. 2023; SGLang's RadixAttention, Zheng et al. 2024). LMCache and CacheGen store, compress and stream caches across tiers and machines (Liu et al. 2024). CacheBlend reuses non-prefix caches and recomputes a selected subset of tokens to repair cross-attention (Yao et al. 2024). Its selective-recompute step is close in mechanics to our column check, but it is used for quality, not integrity. None of these systems verifies a received cache against its tokens.

Attacks on caches. Prior work has studied KV caches as a side channel: prompts leak through the timing of shared prefix caches (Song et al. 2024; Wu et al. 2025). Fault-injection work on LLMs (bit flips in weights, e.g. rowhammer-style attacks) targets parameters, not activations. We look at integrity rather than confidentiality, at the cache rather than the weights, and at a defence that needs no trust in the producer.

Verifiable inference. Zero-knowledge and interactive proofs for neural network inference (e.g. zkLLM, Sun et al. 2024) give cryptographic soundness at a large prover cost. Spot-checking and replication

schemes for untrusted compute (e.g. TOPLOC, Ong et al. 2025, which hashes activations to detect model or precision swaps) trade guarantees for cost. Our check sits at the cheap end: it uses the structure of autoregressive caches (each column is a one-token function of its prefix) to make spot checks exact up to floating-point noise, and attention to choose where to check.

11. Limitations

- **Scale.** Two small models (0.5B and 1.2B parameters), memo-style factual QA with one-token-ish answers, and documents up to ~2K tokens (the attack backpropagates through the whole prefill, which did not fit in 8 GB at ~4K; the audit itself was timed up to 16K). Larger models have more layers and heads, so influence may concentrate differently.
- **Known question.** The attacker optimises for a question it anticipates, and our audit reads the influence of the question actually asked. §7.2 shows that audits driven by other questions do not catch a stealth forgery. We did not test attackers that must succeed on several questions at once, or questions the attacker did not anticipate.
- **Determinism.** The tolerance τ has to be calibrated for each pair of prefill and verification stacks (§7.1). We measured one pair, transformers SDPA against our own kernels, on one GPU. Heterogeneous hardware, quantised caches or different attention kernels would each need their own calibration, and τ may grow.
- **Adaptive attacker.** Our adaptive forgers are white-box but simple: one hinge loss per score and 100–150 Adam steps. Stronger optimisers will hide better. §6 should be read as a lower bound on what an adaptive attacker can do.
- **Not a MAC.** The check proves that a cache is consistent with its tokens and weights. It says nothing about whether the tokens themselves are the right ones. That is the ordinary job of authenticating the prompt.

12. Conclusion

A KV cache can be checked against its tokens column by column, for the price of a few tokens. Every forgery in our experiments fails at the column where it was made, and honest caches pass. Where to check is the hard part. Against faults and non-adaptive tampering, the columns the question reads most are where the problem is: a few columns catch most single-column forgeries, at any document length. Against a white-box attacker that knows the rule, any single score can be gamed. Tracing the audit through the cache, following what the read columns read, makes the attacker's job harder on both models we tested, but only random columns and full recompute give guarantees. That is a real trade-off, and serving systems that move caches between machines can now make it with numbers.

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